AGENT BUILD GUIDE | PRODUCTION WEBSITE

BYOL Academy Website Build Specification

A step-by-step, testable guide for an AI coding agent to design and ship the byolacademy.com homepage and supporting pages on a modern JavaScript stack (Next.js, TypeScript, Tailwind CSS)

Derived from: BYOL Academy competitive analysis (23 October 2026)
Guide date: 25 October 2026
Audience: coding agent + the intern supervising it

READ FIRST

1. How to use this guide

This document is written to be given to a coding agent (for example Claude Code) together with the repository. It tells the agent *what* to build, *why* (traced to the competitive analysis), *how* (stack, structure, code patterns) and *when it is done* (acceptance criteria). Every requirement has an ID (R1, R2...) so work, commits and reviews can reference it.

Operating rules for the agent

- **Work in small phases** (Section 15). Finish a phase, run its checks, commit, then continue. Never leave the build broken.
- **Never invent facts.** Use only the approved content in Section 3. Anything marked TODO(approval) must stay a placeholder until a human approves it. No fake statistics, testimonials, logos, rankings, certifications or legal text.
- **Verify, do not assume.** Package versions, framework APIs and CLI flags change. Before using a tool, check its current docs or `--help`; pin exact versions after install.
- **Accessibility and performance are acceptance criteria**, not polish. A phase is not done if axe or Lighthouse budgets fail.
- **Ask when blocked.** If brand assets, API keys or decisions are missing, stop and list the questions (Section 16). Do not guess brand colours, logos or legal copy.
- **Do not copy competitors.** Use their patterns for inspiration only: no text, images, illustrations or code.
- **Keep it maintainable:** content in typed files, design tokens in CSS variables, small composable components, no dead code.

Master prompt (paste at the start of the agent session)

```
You are the lead front-end engineer for byolacademy.com (BYOL Academy - Be Your Own Light), an Indian
AI adaptive-learning company with three products: Medicoplasma, BYOL Schools, BYOL Colleges.
Read BYOL_Academy_Agent_Build_Guide.pdf completely before writing code. Then:
1. Confirm the open questions in Section 16 with me; proceed with documented defaults if I do not answer.
2. Implement phases 0-7 in order. After each phase run: npm run lint, typecheck, test, build, and the
   Playwright + axe suite. Fix failures before moving on. Commit with Conventional Commits.
3. Use ONLY copy from content/site.ts. Items flagged approved:false must render as clearly marked
   placeholders in development and be hidden in production builds.
4. Honest-AI rule: never claim guaranteed results, ranks, certifications, or that the demo is the
   production engine. The demo is an illustrative simulation and must say so on screen.
5. At the end, produce a LAUNCH_CHECKLIST.md with the status of every requirement R1-R15.
```

FROM THE COMPETITIVE ANALYSIS

2. Context: what we learned and what it requires

BYOL Academy builds one adaptive learning engine that powers **Medicoplasma** (NEET-PG/UKMLE/USMLE prep), **BYOL Schools** (Class 6-12) and **BYOL Colleges** (degrees, diplomas, exams such as SSC). The current homepage tells but does not show, offers only a 'Request a Demo' CTA, hides the products from the navigation, and has thin proof. The redesign must fix these while keeping the brand's honest tone.

ID	Requirement	Traced to finding	Primary section
R1	Make adaptivity visible with an interactive 'See it adapt' demo (illustrative, clearly labelled)	Homepage has no product visual or demo	Section 7: S6
R2	Route each audience (medical aspirant, school student/parent, college student, institution) within 10 seconds	Mixed audience, no routing	S3
R3	Two clear CTA paths: 'Explore platforms' (learners) and 'Request a demo' (institutions)	Only one institutional CTA	Hero, S12

ID	Requirement	Traced to finding	Primary section
R4	Navigation exposes the three platforms, How it works, Research, Blog, About, Log in (if URL provided), CTA button	Products absent from nav	Header
R5	Proof: 20 institutions, 13k+ learners, institution logos (when supplied), testimonials with role and platform tag, incl. a medical learner when approved	Thin proof, no logos, no medical testimonial	S7
R6	'What our AI does and does not do' + responsible-AI principles, with the About page's honest disclaimers	Trust and honesty as differentiator	S8
R7	Working Privacy Policy and Terms routes (content from legal), cookie/consent behaviour matching analytics	Footer links were '#' placeholders	Footer, /privacy, /terms
R8	Mobile-first, fast on mid-range phones and 4G (budgets in Section 9)	Many Indian users on phones (pattern)	All
R9	WCAG 2.2 AA conformance for all pages	Minors and institutions	All
R10	SEO foundation: metadata, Open Graph, sitemap, robots, JSON-LD	Discoverability	Section 9
R11	Privacy-respecting analytics with an event plan and no PII in events	Measure the redesign	Section 11
R12	Secure lead capture: validation, spam control, rate limiting, secrets server-side	Two intents in one form	Section 8
R13	Content governance: typed content with approval flags; nothing unapproved ships	Claims must be team-approved	Section 3
R14	Maintainable codebase: tokens, typed content, tests, CI	Production-ready	Sections 5, 12, 13
R15	i18n-ready structure (no hard-coded strings in components); Hindi deferred pending decision	Vernacular demand (pattern)	Section 5

WHAT MAY BE SHIPPED

3. Content inventory and governance

All user-facing text lives in `src/content/` as typed objects. Each fact-like item has `approved: boolean`. In production builds, items with `approved:false` must not render (the component returns null or a neutral fallback); in development they render with a visible 'NEEDS APPROVAL' badge.

A. Verified content that exists on byolacademy.com today (reusable)

Item	Value
Brand	BYOL Academy - Be Your Own Light; tagline 'Educate - Uplift - Empower'
Announcement	'Launched: Medicoplasma - NEET-PG/UKMLE/USMLE exam prep' linking to medicoplasma.com
Platforms and URLs	Medicoplasma: https://medicoplasma.com | BYOL Schools: https://schools.byolacademy.online | BYOL Colleges: https://schools.byolacademy.online/college
Hero text	'BYOL Adaptive Engine for All Learners.' + description; chips: Personalized Learning Paths, Real-time Progress Insights, Built for Every Learner
Value section	'Learning That Responds - Not Just Records' with four points: Understands the Learner, Precision Over Repetition, Insight in Real Time, Built for Every Stage
Numbers	20 institutions; 13k+ learners (as published on the current homepage)
Engine loop (About)	Learner Evidence > Knowledge Estimation > Personalized Intervention > New Evidence > Updated Learning Path
Principles (About)	Adaptation over standardization; Evidence over assumptions; Domain-specific intelligence; Transparency by design; Privacy and responsible data use; Continuous evaluation
Disclaimers to preserve	Engine is a 'conceptual design, not a description of deployed models'; responsible-AI items are 'principles we design toward, not claims of regulatory compliance'

Item	Value
Testimonials (4)	Dr Manisha Chauhan (MD Psychiatry, BYOL Schools); Sunita Mishra (parent, Class 8, BYOL Schools); Anika Singh (B.Sc. Agriculture and SSC aspirant, BYOL College); Himanshu Singh (M.Sc. Maths and SSC aspirant, BYOL College). Reuse only with confirmed permission and use the existing approved photos/text exactly.
Footer	Platforms, social links (X, LinkedIn, YouTube, Facebook, Instagram), About, Contact, Privacy, Terms; (c) 2026 BYOL Academy

B. Items that must stay TODO(approval) until a human confirms

- Medicoplasma specifics from a startup-portal listing (self-reported): July 2025 launch, 50,000+ question bank, AI mentor 'Dr. Neeto', 2.85 lakh aspirants a year.
- Any claim about results, ranks, score improvement, security certifications, data residency, compliance or 'enterprise-grade' security.
- Institution logos and names (need written permission). Pricing, plans or free trial terms. Response-time promises ('we reply in 24 hours').
- Legal text for Privacy and Terms: obtain from the team; never generate legal text.

C. Content model (TypeScript)

```
// src/content/types.ts
export type Approvable<T> = T & { approved: boolean; note?: string };

export type Platform = Approvable<{
  id: "medicoplasma" | "schools" | "colleges";
  name: string; audience: string; summary: string;
  features: string[]; href: string; ctaLabel: string; external: true;
}>;
export type Testimonial = Approvable<{
  id: string; platform: "schools" | "colleges" | "medicoplasma";
  quote: string; name: string; role: string; city?: string; photo?: string;
}>;
export type Stat = Approvable<{ id: string; value: string; label: string }>;
export type FaqItem = Approvable<{ id: string; q: string; a: string }>;

// helper used by every section
export const shippable = <T extends { approved: boolean }>(xs: T[]) =>
  process.env.NODE_ENV === "production" ? xs.filter((x) => x.approved) : xs;
```

JAVASCRIPT / TYPESCRIPT, PRODUCTION-READY

4. Technology stack

Evidence suggests the current site uses Next.js and Vercel storage (inferred from page metadata and image URLs). The stack below matches that so work can merge into their codebase. Confirm with the dev team (Section 16). **Versions:** install the latest stable releases, then pin exact versions in package.json and commit the lockfile.

Layer	Choice	Why / notes
Framework	Next.js (App Router), React, TypeScript strict	SSR/SSG, metadata API, image and font optimisation, route handlers, matches probable existing stack
Styling	Tailwind CSS (CSS-first config with @theme tokens)	Fast, consistent, small CSS output
Components	shadcn/ui (Radix primitives) copied into the repo	Accessible dropdowns, accordion, dialog, form parts; you own the code
Icons	lucide-react	Tree-shakeable, consistent
Motion	motion (the Framer Motion library, current package name: verify) with reduced-motion support	Engine loop and demo transitions; keep subtle

Layer	Choice	Why / notes
Visualisation	Custom SVG for the mastery map; Recharts only if a chart is needed	Smallest bundle; full control of accessibility
Forms	React Hook Form + Zod (+ @hookform/resolvers)	Shared client/server validation
Email / leads	Resend (or the team's CRM webhook, TBD) via a route handler	Server-side secret; swap-able adapter
Spam / abuse	Honeypot + rate limit (Upstash Ratelimit/Redis) + optional Cloudflare Turnstile	Protects the demo form
Analytics	Vercel Web Analytics and/or a privacy-friendly tool (Plausible, PostHog) behind one wrapper	Swap vendors without touching components
Monitoring	Sentry (errors) + Vercel logs	Production visibility
Testing	Vitest + Testing Library (unit), Playwright + @axe-core/playwright (e2e and a11y), Lighthouse CI	Automated gates for R8, R9
Quality	ESLint (next config), Prettier + tailwind plugin, Husky + lint-staged, Conventional Commits	Consistent code
Hosting	Vercel (preview per PR, production on main)	Zero-config for Next.js; confirm the team's hosting
Package manager	npm (or pnpm if the team uses it)	One lockfile only

Avoid
Runtime CSS-in-JS, heavy carousel/animation libraries, jQuery, large icon packs imported wholesale, client-side-only rendering of marketing content, unvetted copy-pasted snippets, any library that cannot be justified against the JS budget in Section 9.

PHASE 1 REFERENCE

5. Project setup and architecture

```
npx create-next-app@latest byol-academy-web --typescript --tailwind --eslint --app --src-dir \
  --import-alias "@/*"          # run with --help first; flags can change between versions
cd byol-academy-web
npx shadcn@latest init           # choose neutral base; CSS variables: yes
npx shadcn@latest add button card navigation-menu accordion form input label select \
  textarea sheet badge tabs checkbox
npm i motion zod react-hook-form @hookform/resolvers resend @upstash/ratelimit @upstash/redis
npm i -D vitest @vitejs/plugin-react jsdom @testing-library/react @testing-library/jest-dom \
  @playwright/test @axe-core/playwright prettier prettier-plugin-tailwindcss @lhci/cli husky lint-staged
npx playwright install --with-deps chromium
```

Folder structure

```
src/
  app/
    layout.tsx            # fonts, metadata base, skip link, header/footer, analytics provider
    page.tsx              # homepage: composes section components
    about/page.tsx  contact/page.tsx  blog/(index + [slug])  privacy/page.tsx  terms/page.tsx
    api/lead/route.ts     # POST lead capture (validated, rate-limited)
    sitemap.ts  robots.ts  opengraph-image.tsx  not-found.tsx  error.tsx
  components/
    ui/                   # shadcn primitives (Button, Card, Accordion...)
    layout/               # Header, Footer, MobileNav, SkipLink, AnnouncementBar
    sections/             # Hero, AudienceChooser, PlatformCards, EngineLoop, AdaptDemo,
                          # Proof, TrustAI, ForInstitutions, Faq, ClosingCta, BlogTeaser
    demo/                 # QuestionCard, MasteryMap, DemoSummary, useDemoMachine.ts
    forms/                # LeadForm, EmailCapture
  content/                # site.ts, platforms.ts, testimonials.ts, faq.ts, demo-data.ts, types.ts
  lib/
```

```
    adaptive/mastery.ts   # illustrative BKT update (unit-tested)
    analytics.ts  validation/lead.ts  ratelimit.ts  cn.ts  seo.ts
  styles/globals.css      # Tailwind import + @theme tokens
tests/  unit/  e2e/       # vitest and playwright
.github/workflows/ci.yml
.env.example   lighthouserc.json   next.config.ts   LAUNCH_CHECKLIST.md
```

Configuration rules

- `tsconfig.json`: `"strict": true`, `noUncheckedIndexedAccess: true`. No any without a comment.
- Server Components by default. Add `"use client"` only to interactive leaves (nav menu, demo, form, accordion trigger wrappers).
- Environment variables: only `NEXT_PUBLIC_*` values may reach the browser. Commit `.env.example`; never commit secrets.
- Env keys: RESEND_API_KEY, LEAD_TO_EMAIL, LEAD_FROM_EMAIL, UPSTASH_REDIS_REST_URL, UPSTASH_REDIS_REST_TOKEN, NEXT_PUBLIC_SITE_URL, NEXT_PUBLIC_ANALYTICS_ID, SENTRY_DSN, TURNSTILE_SECRET (optional).
- Strings: components import copy from `src/content`; no literal marketing text in JSX (R15).

TOKENS, TYPE, COMPONENTS

6. Design system

Brand status
I could not retrieve the live site's colours or fonts. The palette below is a **proposal** (warm amber for 'light', deep navy for trust). If the team supplies brand guidelines, change only the token values; components must not hard-code colours.

A. Colour tokens and verified contrast

Token pair	Foreground	Background	Contrast (WCAG AA needs 4.5:1)
Body text	#0B1B3B	#FFFFFF	17.0:1
Body text on canvas	#0B1B3B	#FAF7F0	15.9:1
Muted text	#4A5470	#FFFFFF	7.5:1
Muted text on canvas	#4A5470	#FAF7F0	7.0:1
Button label on primary	#FFFFFF	#1F3A8A	10.3:1
Ink on accent (CTA)	#0B1B3B	#F5B301	9.2:1
Primary as link/text	#1F3A8A	#FFFFFF	10.3:1
Ink on accent-soft	#0B1B3B	#FFF4D6	15.5:1
Success text	#0F7B55	#FFFFFF	5.3:1
Warning text	#9A3F00	#FFFFFF	6.8:1
Error text	#B42318	#FFFFFF	6.6:1
White on ink (footer)	#FFFFFF	#0B1B3B	17.0:1

Contrast ratios above were computed with the WCAG relative-luminance formula when this guide was generated. Rule: accent amber (#F5B301) is a **background or graphic colour only**, never text on white.

```
/* src/styles/globals.css (Tailwind CSS-first; verify @theme syntax against current docs) */
@import "tailwindcss";
@theme {
  --color-ink: #0B1B3B;        --color-canvas: #FAF7F0;     --color-surface: #FFFFFF;
  --color-primary: #1F3A8A;    --color-primary-700: #172B66;
  --color-accent: #F5B301;     --color-accent-soft: #FFF4D6;
  --color-muted: #4A5470;      --color-border: #E3DDCF;
  --color-success: #0F7B55;    --color-warning: #9A3F00;    --color-danger: #B42318;
```

```
  --font-sans: var(--font-inter), system-ui, sans-serif;
  --font-display: var(--font-jakarta), var(--font-inter), system-ui, sans-serif;
  --radius-card: 16px;  --radius-btn: 12px;
  --shadow-card: 0 1px 2px rgb(11 27 59 / 0.06), 0 8px 24px rgb(11 27 59 / 0.06);
}
:focus-visible { outline: 3px solid var(--color-primary); outline-offset: 2px; }
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after { animation-duration: 0.01ms !important; transition-duration: 0.01ms !important; }
}
```

B. Typography, spacing, layout

Token	Spec
Fonts	Headings: Plus Jakarta Sans; body: Inter, both via `next/font/google` with `display: swap` and Latin subset. If Hindi is approved later, add Noto Sans Devanagari the same way.
Scale (px)	Body 16/24 (18/28 on desktop for lead text); H3 20/28; H2 32/40 (mobile 26/34); H1 56/60 desktop, 36/42 mobile. Use clamp() for fluid headings.
Spacing	8-point scale: 4, 8, 12, 16, 24, 32, 48, 64, 96. Section padding: 64px mobile, 96px desktop.
Layout	Max content width 1200px, centred; 12-column grid desktop (24px gutter), 4-column mobile (16px margin). Breakpoints: 390, 768, 1024, 1440.
Radius and elevation	Cards 16px, buttons 12px, one soft shadow level; borders 1px token border colour.
Motion	Durations 150ms (micro), 300ms (section), easing ease-out. Only transform and opacity animate. All motion disabled under prefers-reduced-motion.
Iconography	Lucide, 20/24px, stroke 1.75; decorative icons aria-hidden; meaningful icons have labels.

C. Component inventory and props

Component	Props (summary)	Behaviour and a11y
Button	variant: primary | secondary | ghost; size; asChild; loading	Primary = accent bg + ink text; min height 44px; visible focus; disabled uses aria-disabled
Header / MobileNav	items: NavItem[]; cta	Platforms dropdown via Radix NavigationMenu; mobile uses Sheet with focus trap; sticky; skip link first in DOM
AnnouncementBar	text, href, dismissible	Dismiss stored in sessionStorage; role='region' aria-label='Announcement'
Section	id, eyebrow, title, intro, tone	Wraps every section with correct heading level and aria-labelledby
AudienceTile	audience, title, blurb, href, icon	Whole tile is one link; external link adds visually hidden '(opens in same tab on another BYOL site)'
PlatformCard	platform: Platform	Colour-tagged by platform plus icon and text label (never colour alone)
EngineLoop	steps[4 or 5]	Static ordered list in markup, SVG ring as decoration; animation optional
AdaptDemo	dataset, onComplete	State machine; keyboard operable; aria-live results; text alternative table (Section 7)
MasteryMap	concepts, mastery, edges	SVG nodes with text labels and status icons; hidden table equivalent
StatCounter	stat: Stat	Static number (no count-up for reduced-motion users)
TestimonialCard	testimonial	Blockquote + cite; image alt = person name; no auto-rotating carousel
FaqAccordion	items	Radix Accordion; one open at a time optional; JSON-LD FAQPage only if answers approved
LeadForm / EmailCapture	intent, onSuccess	Labels, inline errors, aria-describedby, honeypot, submit state, success message region
Footer	columns, social, legal	Real links (no '#'); social links have accessible names

PHASE 3 AND 4

7. Homepage specification, section by section

Order and intent come from the wireframe in the analysis. Copy shown is **draft** and built from approved site text; the human must approve final wording. Each section lists layout, content, behaviour, analytics and acceptance.

S0-S1. Announcement bar and header (R4)

Aspect	Specification
Layout	Bar 40px; header 72px desktop, 64px mobile, sticky with subtle border on scroll.
Content	Bar: 'Launched: Medicoplasma - NEET-PG/UKMLE/USMLE exam prep' + 'Learn more'. Nav: Platforms (Medicoplasma, BYOL Schools, BYOL Colleges), How it works (#how-it-works), Research (/about#research), Blog, About, Contact; right side: Log in (only if a URL is supplied), primary button 'Request a demo'.
Behaviour	Dropdown opens on click/Enter, closes on Esc; mobile menu in a sheet; header never covers anchored content (scroll-margin-top).
Events	nav_click {item}; cta_click {id:'header_demo'}.
Accept	Keyboard-only user can open Platforms and reach every link; axe: 0 violations.

S2. Hero (R1, R3)

Aspect	Specification
Layout	Desktop: 7/5 split, text left, **MasteryMap preview** right. Mobile: text first, visual below, CTAs full width.
Content (draft)	Eyebrow: 'AI-powered learning platform'. H1: 'Learning that responds to you.' Sub: 'The BYOL Adaptive Engine meets every learner where they are and helps decide what to practise next, across medicine, schools and colleges.' CTAs: primary 'Explore platforms' (#platforms), secondary 'Request a demo' (/contact?intent=institution). Chips: Personalized Learning Paths, Real-time Progress Insights, Built for Every Learner. Trust line: 'Used by 20 institutions and 13k+ learners.'
Visual	Static SVG mastery map with 6 labelled concepts (strong, shaky, missing) and a caption 'Illustrative example'. Gentle node pulse only if motion allowed.
Performance	H1 is the LCP element; no hero image above 60 KB; no layout shift; visual is inline SVG.
Events	cta_click {id:'hero_explore'|'hero_demo'}.
Accept	LCP under 2.5s on Lighthouse mobile preset; CTAs distinguishable without colour.

S3. Audience chooser (R2)

Aspect	Specification
Layout	Heading 'Where are you on your learning journey?' 4 tiles (2x2 mobile, 4x1 desktop).
Tiles	Medical aspirant > medicoplasma.com; School student or parent > schools.byolacademy.online; College student or exam aspirant > schools.byolacademy.online/college; School or institution > /contact?intent=institution.
Behaviour	Each tile is one link with hover/focus lift; no JS needed.
Events	audience_select {audience}.
Accept	Every tile reachable and announced with its destination; tap target at least 44px.

S4. Platform cards (R3, R4)

Aspect	Specification
Layout	Anchor id='platforms'. Three equal cards; each: icon, name, audience line, summary, 3 features, CTA.
Medicoplasma	Summary: 'Adaptive assessment and preparation for medical students and professionals.' Features (approved wording only): adaptive assessment; structured learning and practice; domain-specific knowledge development. CTA 'Explore Medicoplasma'. Question-bank size stays TODO(approval).

Aspect	Specification
BYOL Schools	Summary: 'Personalized learning for every student, from Class 6 to 12.' Features: AI-powered doubt-solving; adaptive testing; measurable progress across subjects. CTA 'Explore BYOL Schools'.
BYOL Colleges	Summary: 'Adaptive preparation for degrees, diplomas, and the exams that follow them.' Features: understands your course and goals; focuses on what to study next; doubt support (BYOL SAARTHI, per testimonials; confirm naming). CTA 'Explore BYOL Colleges'.
Events	platform_card_click {platform}.
Accept	Cards equal height; external destination indicated; copy matches content files exactly.

S5. How the engine works (R6)

Aspect	Specification
Content	Heading 'Learning That Responds - Not Just Records'. Steps from the About page: 1 Learner Evidence, 2 Knowledge Estimation, 3 Personalized Intervention, 4 New Evidence, 5 Updated Learning Path (loop). Short line under each from About copy.
Layout	Horizontal loop on desktop, vertical list on mobile; markup is an ordered list; SVG arrows decorative.
Footnote	'Conceptual architecture. Implementation details evolve through research and product validation.' (approved About wording).
Accept	Meaningful without CSS; passes reading-order check.

S6. 'See it adapt' demo (R1) - full spec in Section 8

Aspect	Specification
Layout	Card with three steps: pick a subject, answer 3 questions, see your mastery map and a suggested next step.
Labelling	Visible, persistent label: 'Illustrative simulation with sample questions. Not your real results and not the production engine.'
Events	demo_start, demo_answer, demo_complete {subject, correctCount}.
Accept	Complete flow works by keyboard and screen reader; no network calls; under 30 KB gzip added JS.

S7. Proof (R5)

Aspect	Specification
Content	Stats: 20 institutions, 13k+ learners. Logo wall renders only when logos are supplied with permission (otherwise hidden). Testimonials: 4 existing, each with platform tag, role and city; add a medical learner when approved.
Layout	Stats row; testimonial grid 2x2 (no autoplay carousel); photo 64px circle with alt text.
Accept	Production hides unapproved items; no layout break when logo wall is empty.

S8. Trust and responsible AI (R6)

Aspect	Specification
Content	Two columns. 'What our AI does': estimates understanding from learner responses; suggests what to practise next; shows progress. 'What it does not do': replace teachers or experts; make final decisions about a learner; (add only team-approved statements). Below: six principles from the About page as cards, then the exact disclaimer: 'These are principles we design toward. They are not claims of regulatory compliance or certification.'
Links	'Read our principles' > /about#responsible-ai; Privacy Policy link.
Accept	No security or compliance claim appears unless flagged approved.

S9. For institutions (R3)

Aspect	Specification
Content	'For schools and institutions': real-time progress insights, concept-level visibility, adaptive testing (only items present in approved copy). CTA 'Request a demo'. Optional dashboard illustration (static, labelled illustrative).
Accept	CTA goes to /contact?intent=institution and pre-selects the intent field.

S10-S11. FAQ and blog teaser

Aspect	Specification
FAQ draft topics	What is adaptive learning? How is BYOL different from a test that only gives a score? Which learners is it for? How do I try it? How do schools get started? How is learner data handled? (answers must be team-approved; show only approved ones).
Blog	Latest 3 posts from the blog source (MDX or the team's CMS); skip the section if none.
Accept	Accordion keyboard-operable; FAQPage JSON-LD only for approved answers.

S12. Closing CTA and footer (R3, R7)

Aspect	Specification
Content	'The future of learning adapts to you.' Two paths side by side: Learners > 'Explore platforms'; Institutions > 'Request a demo' / 'Talk to our team'. Optional EmailCapture ('Get started' with intent select learner or institution).
Footer	Platforms, Resources (X, LinkedIn, YouTube, Facebook, Instagram), Company (About, Contact, Privacy, Terms), copyright 2026 BYOL Academy.
Accept	Privacy and Terms resolve to real routes; each social link has an accessible name and rel='noopener noreferrer'.

PHASE 4 REFERENCE

8. 'See it adapt' demo: engine, data and states

Purpose: let visitors *feel* adaptivity in under 30 seconds. **Honesty constraint:** the About page states the engine is a conceptual design, so the demo is an **illustrative simulation** using sample data and a textbook Bayesian Knowledge Tracing (BKT) update. It must never be described as BYOL's production model.

A. Flow and states

State	UI	Transition
idle	Subject picker: Medicine, School maths, College quantitative aptitude (tabs/radio group). Start button.	start > asking
asking	One question at a time, 4 options, progress 'Question 2 of 3', concept tag shown.	select option > feedback
feedback	Correct/incorrect with a one-line explanation; MasteryMap updates (node estimate changes); next button.	next > asking or summary
summary	Final map; list sorted by lowest mastery; suggested next step ('Practise: ...'); reset and 'Explore the real platform' CTA.	reset > idle

B. Data model and sample content

```
// src/content/demo-data.ts  (ALL sample data; label it as illustrative in the UI)
export type Concept = { id: string; label: string; prior: number; prereqs: string[] };
export type Question = {
  id: string; conceptId: string; prompt: string;
  options: string[]; correctIndex: number; explanation: string;
};
export type DemoSubject = {
  id: "medicine" | "school-maths" | "college-quant";
  label: string; concepts: Concept[]; questions: Question[];   // exactly 3 questions shown
};
// Example (school maths) - authors must double-check every answer before shipping:
// concepts: fractions, ratios, linear-equations, word-problems; priors ~0.3-0.5
// Q1 ratios: "If 3 pens cost 45 rupees, what do 5 pens cost?" options 60/75/90/105, correct 75.
```

Content safety
Questions must be factually correct and reviewed by a subject expert. For the Medicine subject, use only basic, uncontroversial physiology questions with no clinical advice, and add: 'For learning illustration only; not medical guidance.' Never use real learner data.

C. Mastery update (unit-tested)

```
// src/lib/adaptive/mastery.ts - textbook Bayesian Knowledge Tracing, illustrative only
export type BktParams = { slip: number; guess: number; learn: number };
export const DEFAULTS: BktParams = { slip: 0.1, guess: 0.2, learn: 0.15 };

export function updateMastery(p: number, correct: boolean, k: BktParams = DEFAULTS): number {
  const { slip, guess, learn } = k;
  const posterior = correct
    ? (p * (1 - slip)) / (p * (1 - slip) + (1 - p) * guess)
    : (p * slip) / (p * slip + (1 - p) * (1 - guess));
  const next = posterior + (1 - posterior) * learn;
  return Math.min(0.99, Math.max(0.01, next));
}

export type Status = "strong" | "developing" | "needs-work";
export const statusOf = (p: number): Status =>
  p >= 0.75 ? "strong" : p >= 0.4 ? "developing" : "needs-work";

// Suggest the lowest-mastery concept whose prerequisites are all >= 0.4; fall back to lowest overall.
export function nextStep(m: Record<string, number>, prereqs: Record<string, string[]>): string {
  const ids = Object.keys(m).sort((a, b) => m[a]! - m[b]!);
  return ids.find((id) => (prereqs[id] ?? []).every((q) => (m[q] ?? 0) >= 0.4)) ?? ids[0]!;
}
```

Unit tests required

- A correct answer never decreases mastery. From the same starting value, mastery after a correct answer is always higher than after an incorrect one. With learn set to 0, an incorrect answer never increases mastery (with a non-zero learn rate it can rise slightly from very low values; this is expected BKT behaviour).
- Output always within [0.01, 0.99]. `statusOf` thresholds at 0.4 and 0.75. `nextStep` respects prerequisites and has a fallback.

D. Visual and accessibility requirements

- MasteryMap is an SVG with 4-6 labelled nodes; each node shows text label, percentage, **and** an icon/shape per status (check, half, exclamation). Colour is supplementary (green, amber-brown, red tokens meet contrast).
- A visually hidden `<table>` lists concept, mastery %, status for screen readers. Updates announced in an `aria-live='polite'` region (e.g. 'Ratios: now developing, 52 percent').
- Options are a radio group with arrow-key support; Enter/Space selects; focus moves to the feedback heading after answering, then to Next.
- Respect reduced motion: no node animation; values change instantly.
- No network requests, no storage of answers, no personal data. Analytics events carry only subject and counts.

R8 AND R10

9. Performance, SEO and structured data

Budget	Target	How to meet it
Largest Contentful Paint	under 2.5s (mobile, slow 4G profile)	Text LCP, inline SVG hero visual, next/font, no render-blocking scripts
Interaction to Next Paint	under 200ms	Server Components, small client islands, avoid long tasks
Cumulative Layout Shift	under 0.1	Reserve image/logo sizes; font-display swap with size-adjust (next/font does this)
Total Blocking Time (lab)	under 200ms	Defer analytics; lazy-load demo with dynamic import below the fold
First-load JS (homepage)	aim at or below about 150 KB gzip (team guideline, adjust if needed)	Track with the Next build output and @next/bundle-analyzer
Images	WebP/AVIF via next/image, sizes attribute, lazy below fold; each testimonial photo under 20 KB	Existing photos are 192px WebP; keep that
Lighthouse (mobile)	Performance 90+, Accessibility 95+, Best Practices 95+, SEO 95+	Enforced in CI (Section 13)

SEO implementation

- Metadata API in layout and each page: title template, description, canonical, `openGraph` (site name BYOL Academy, locale en_IN), Twitter card, reuse the existing `opengraph-image` approach.
- `app/sitemap.ts` and `app/robots.ts` (disallow /api). Block indexing of preview deployments.
- JSON-LD on the homepage: `Organization` (name, url, logo, sameAs social links) and `WebSite`. Add `FAQPage` only for approved FAQ items. Render JSON-LD in a script tag with `type='application/ld+json'`.
- One H1 per page; meaningful link text ('Explore BYOL Schools', not 'Learn more' alone); descriptive alt text; lang='en-IN' on html.
- Lead copy for search: 'adaptive learning', 'AI-powered learning platform', 'NEET-PG preparation', 'personalised learning for Class 6-12' only where approved copy already uses them.

R7 AND R12

10. Lead capture, security and privacy

Forms: **LeadForm** on /contact (fields: name, work or personal email, role/intent [learner, parent, institution, other], organisation (required when institution), message, consent checkbox) and optional **EmailCapture** (email + intent). The marketing site must be aimed at adults (parents, educators, institutions, learners 18+). Do not collect data from children here. Because India's Digital Personal Data Protection Act, 2023 and its rules govern personal data (including special treatment of children's data), have the team's legal reviewer approve the consent wording and privacy notice before launch.

```
// src/lib/validation/lead.ts
import { z } from "zod";
export const leadSchema = z.object({
  name: z.string().trim().min(2).max(100),
  email: z.string().trim().email().max(200),
  intent: z.enum(["learner", "parent", "institution", "other"]),
  organisation: z.string().trim().max(150).optional(),
  message: z.string().trim().max(2000).optional(),
  consent: z.literal(true),                    // explicit consent required
  website: z.string().max(0).optional(),       // honeypot: must be empty
```

```
}).refine((v) => v.intent !== "institution" || !!v.organisation, {
  path: ["organisation"], message: "Please tell us your institution",
});
export type LeadInput = z.infer<typeof leadSchema>;

// src/app/api/lead/route.ts (server only; secrets never reach the client)
import { NextResponse } from "next/server";
import { Resend } from "resend";
import { leadSchema } from "@/lib/validation/lead";
import { limiter } from "@/lib/ratelimit";

export async function POST(req: Request) {
  const ip = (req.headers.get("x-forwarded-for") ?? "unknown").split(",")[0]!.trim();
  const { success } = await limiter.limit(ip);                // e.g. 5 requests / 10 minutes
  if (!success) return NextResponse.json({ ok: false }, { status: 429 });

  const body = await req.json().catch(() => null);
  const parsed = leadSchema.safeParse(body);
  if (!parsed.success)
    return NextResponse.json({ ok: false, errors: parsed.error.flatten().fieldErrors }, { status: 400 });
  if (parsed.data.website) return NextResponse.json({ ok: true });   // bot: pretend success

  const resend = new Resend(process.env.RESEND_API_KEY);
  const { name, email, intent, organisation, message } = parsed.data;
  await resend.emails.send({
    from: process.env.LEAD_FROM_EMAIL!, to: process.env.LEAD_TO_EMAIL!, replyTo: email,
    subject: `New ${intent} enquiry from ${name}`,
    text: `Name: ${name}\nEmail: ${email}\nIntent: ${intent}\nOrg: ${organisation ?? "-"}\n\n${message ?? ""}`,
  });
  return NextResponse.json({ ok: true });
}
```

Security checklist

Control	Requirement
Security headers	Set in next.config.ts: Strict-Transport-Security, X-Content-Type-Options: nosniff, Referrer-Policy: strict-origin-when-cross-origin, Permissions-Policy (disable camera, microphone, geolocation), X-Frame-Options or frame-ancestors 'none'.
Content Security Policy	Start in report-only mode, then enforce. Prefer nonce-based CSP via middleware (renamed 'proxy' in newer Next.js versions: check the docs). Allow only own origin, the analytics host and Vercel image hosts.
Input handling	Validate on server with Zod; never trust client validation; escape output; plain-text emails only.
Secrets	Server env only; rotate on exposure; never log PII or secrets.
Abuse	Honeypot + rate limit (and Turnstile if spam appears); return generic errors.
Dependencies	npm audit in CI; Dependabot or Renovate; lockfile committed; no abandoned packages.
External links	rel='noopener noreferrer' on target=_blank; clearly mark links to other BYOL domains.
Privacy	Analytics without PII; cookie banner only if non-essential cookies are set; Privacy Policy states what the form collects and why; data retention agreed with the team.

```
// next.config.ts (excerpt) - extend with CSP after testing in report-only mode
const securityHeaders = [
  { key: "Strict-Transport-Security", value: "max-age=63072000; includeSubDomains; preload" },
  { key: "X-Content-Type-Options", value: "nosniff" },
  { key: "Referrer-Policy", value: "strict-origin-when-cross-origin" },
  { key: "Permissions-Policy", value: "camera=(), microphone=(), geolocation=()" },
  { key: "X-Frame-Options", value: "DENY" },
];
const nextConfig = { async headers() { return [{ source: "/(.*)", headers: securityHeaders }]; } };
export default nextConfig;
```

R9 AND R11

11. Accessibility and analytics

Accessibility checklist (WCAG 2.2 AA)

Area	Requirement
Structure	Landmarks (header, nav, main, footer); skip link to #main (exists on current site: keep); one H1; logical heading order; lang='en-IN'.
Keyboard	All interactive elements reachable and operable; visible focus (3px ring); no keyboard traps; Esc closes menus and sheets.
Targets	Interactive targets at least 44x44px (WCAG 2.2 minimum is 24px; this guide uses 44px as the standard).
Colour and contrast	Text 4.5:1 (3:1 for large text and UI components); colour never the only signal; verified token pairs in Section 6.
Content	Alt text for informative images, empty alt for decorative; descriptive links; no auto-playing media; no auto-rotating carousels.
Forms	Programmatic labels, required indicated in text, errors linked with aria-describedby and announced, no placeholder-as-label, autocomplete attributes (name, email).
Motion	prefers-reduced-motion respected; no flashing content.
Dynamic content	aria-live for demo feedback and form status; focus management after state changes.
Testing	Automated axe on every page and state; manual keyboard pass; screen reader smoke test (NVDA with Firefox or Chrome; VoiceOver on iOS; TalkBack on Android).

Analytics event plan (no PII)

Event	Properties	Purpose
cta_click	id (header_demo, hero_explore, hero_demo, closing_learner, closing_institution)	Compare learner vs institution intent
audience_select	audience	Which audience is most common
platform_card_click	platform	Product interest
demo_start / demo_answer / demo_complete	subject, questionIndex, correct (boolean), correctCount	Demo engagement and completion rate
form_start / form_submit_success / form_submit_error	form (lead|email), intent	Conversion funnel
faq_open	id	Which objections matter
scroll_depth	25, 50, 75, 100	Section engagement
outbound_click	destination (platform id)	Handoff to product sites

```
// src/lib/analytics.ts - single wrapper so vendors can be swapped without touching components
type Props = Record<string, string | number | boolean>;
export function track(event: string, props: Props = {}) {
  if (typeof window === "undefined") return;
  // never pass emails, names or free text here
  window.dispatchEvent(new CustomEvent("byol:track", { detail: { event, props } }));
  // adapter: forward to the chosen provider (Vercel Analytics, Plausible, PostHog) in one place
}
```

R8, R9, R14

12. Testing and quality gates

Layer	Tool	What to cover
Unit	Vitest + Testing Library	mastery.ts maths, shippable() filtering, zod schema, content integrity (no empty strings, links valid)
Component	Testing Library	AudienceTile links, FaqAccordion keyboard, LeadForm error states, MasteryMap table equivalent
End to end	Playwright (Chromium, plus WebKit if time allows) at 390px and 1440px	Nav and dropdown; audience tiles; demo complete flow; form success and validation; 404; legal pages exist
Accessibility	@axe-core/playwright	Zero serious or critical violations on all pages and demo states
Performance	Lighthouse CI	Budgets in Section 9 on the production build
Visual	Optional Playwright screenshots	Header, hero, cards, footer at two widths
Manual	Real devices	Low-end Android over throttled 4G, iPhone Safari, 360px width, 200% zoom, keyboard-only, Windows high contrast

```
// tests/e2e/a11y.spec.ts
import { test, expect } from "@playwright/test";
import AxeBuilder from "@axe-core/playwright";

for (const path of ["/", "/about", "/contact", "/privacy", "/terms"]) {
  test(`no serious a11y violations on ${path}`, async ({ page }) => {
    await page.goto(path);
    const results = await new AxeBuilder({ page })
      .withTags(["wcag2a", "wcag2aa", "wcag21aa", "wcag22aa"]).analyze();
    const bad = results.violations.filter((v) => ["serious", "critical"].includes(v.impact ?? ""));
    expect(bad, JSON.stringify(bad, null, 2)).toEqual([]);
  });
}

// tests/e2e/demo.spec.ts - demo works by keyboard
import { test, expect } from "@playwright/test";
test("adapt demo completes", async ({ page }) => {
  await page.goto("/#see-it-adapt");
  await page.getByRole("button", { name: /start/i }).click();
  for (let i = 0; i < 3; i++) {
    await page.getByRole("radio").first().check();
    await page.getByRole("button", { name: /next|see results/i }).click();
  }
  await expect(page.getByText(/illustrative simulation/i)).toBeVisible();
  await expect(page.getByRole("table")).toBeAttached();   // text equivalent of the map
});
```

PHASE 7

13. CI/CD and deployment

```
# .github/workflows/ci.yml
name: ci
on: [pull_request, push]
jobs:
  verify:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: lts/*, cache: npm }
      - run: npm ci
      - run: npm run lint
      - run: npm run typecheck        # tsc --noEmit
      - run: npm test -- --run        # vitest
      - run: npm run build
      - run: npx playwright install --with-deps chromium
      - run: npx playwright test
      - run: npx lhci autorun         # reads lighthouserc.json
```

```
      - run: npm audit --audit-level=high

// lighthouserc.json
{ "ci": {
    "collect": { "startServerCommand": "npm run start", "url": ["http://localhost:3000/"], "numberOfRuns": 3 },
    "assert": { "assertions": {
      "categories:performance":   ["error", { "minScore": 0.9 }],
      "categories:accessibility": ["error", { "minScore": 0.95 }],
      "categories:best-practices":["error", { "minScore": 0.95 }],
      "categories:seo":           ["error", { "minScore": 0.95 }],
      "largest-contentful-paint": ["error", { "maxNumericValue": 2500 }],
      "cumulative-layout-shift":  ["error", { "maxNumericValue": 0.1 }],
      "total-blocking-time":      ["error", { "maxNumericValue": 200 }] } } } }
```

Deployment steps

- Connect the repo to Vercel (or the team's host). Preview deployment for every PR; production from main. Set env vars per environment; never expose server keys.
- Add the domain, enforce HTTPS and the canonical host (www vs apex decision with the team); keep redirects for existing URLs (/about, /contact, /blog).
- Enable Sentry release tracking, uptime monitoring and a rollback plan (promote previous deployment).
- Mark preview deployments noindex. Verify sitemap in Google Search Console after launch.
- Run the launch checklist (Section 17) on the production URL, not only locally.

SCOPE

14. Page inventory and priorities

Priority	Page	Notes
P0	/ (Homepage)	All sections S0-S12; this guide's focus
P0	/contact	LeadForm with ?intent= prefill; success state; alternative email if the team provides one
P0	/privacy and /terms	Routes with content from legal. If not supplied, show 'Content pending legal review' and flag as a launch blocker. Do not write legal text.
P0	404 and error pages	Branded, helpful links to platforms and contact
P1	/about	Restyle existing content with the new design system: vision, engine architecture, research directions, responsible AI, team. Preserve the honest wording. Add anchors #research and #responsible-ai.
P1	/blog and /blog/[slug]	MDX or CMS; article layout with readable measure (65-75 characters), share links, related posts
P2	Log in link	Only if the product sites provide a URL
P2	Hindi / vernacular	Deferred until the team decides; structure already i18n-ready (R15)

EXECUTION ORDER

15. Phased task plan with acceptance criteria

Phase	Tasks	Acceptance criteria (all must pass)
0. Discovery	Confirm open questions (Section 16); request brand assets, logos, approved copy and permissions; confirm stack and hosting	Written answers or recorded defaults; assets folder created; content approval list started
1. Foundation	Scaffold project; TypeScript strict; ESLint, Prettier, Husky; CI pipeline; folder structure; env example; deploy empty app to preview	CI green on a trivial PR; preview URL works; lint, typecheck, build pass
2. Design system	Tokens in globals.css; fonts; Button, Card, Section, Header, Footer, AnnouncementBar; contrast table checked; a /styleguide route (noindex) showing components	All components keyboard-operable; axe clean on /styleguide; tokens only (no hard-coded colours)

Phase	Tasks	Acceptance criteria (all must pass)
3. Static sections	Content files and types; S0-S5, S7-S12 built mobile-first; responsive at 390/768/1440	Matches Section 7; unapproved content hidden in production build; Lighthouse performance 90+ on mobile
4. Demo	mastery.ts + tests; demo data; state machine; MasteryMap with table equivalent; live regions; analytics	Unit and e2e demo tests pass; works with keyboard and screen reader; labelled illustrative; added JS under 30 KB gzip
5. Forms and API	LeadForm, EmailCapture, route handler, validation, rate limit, email adapter, success and error states	Valid submit delivers email in preview; invalid shows linked errors; honeypot and rate limit tested; no secrets client-side
6. SEO, a11y, perf	Metadata, OG, sitemap, robots, JSON-LD; security headers; fix axe and Lighthouse issues; analytics wrapper and consent behaviour	Lighthouse budgets met; axe zero serious/critical; headers verified with a header scanner; sitemap valid
7. QA and launch	Real-device tests; cross-browser; content approval sign-off; legal pages; redirects; monitoring; LAUNCH_CHECKLIST.md; production deploy	All R1-R15 marked done or explicitly waived by a human; rollback documented

ASK BEFORE BUILDING

16. Open questions and default assumptions

Question	Default if unanswered
Brand: official logo files, colours, fonts?	Use the proposed palette and fonts in Section 6 and a text wordmark; flag as provisional
Is the site on Next.js and Vercel today? Who deploys?	Assume yes; deploy previews on Vercel
Primary goal: institution demo requests or learner sign-ups?	Both: dual CTAs with equal prominence; demo requests first in header
Which product is the priority now?	Medicoplasma first in order (it is the announced launch), then Schools, Colleges
Approved numbers and claims for public use?	Use only 20 institutions and 13k+ learners; hide everything else
Is there a Log in URL for the product sites?	Omit Log in until provided
Where should leads go (email, CRM, spreadsheet)?	Email via Resend to a team inbox set in env vars; keep an adapter interface
Legal pages: who provides Privacy and Terms text?	Create routes with 'pending legal review' and mark as launch blocker
Analytics tool preference and cookie policy?	Privacy-friendly analytics without cookies; no banner needed if no non-essential cookies
Hindi or other languages planned?	English (en-IN) only; structure ready for i18n
Permission for testimonial photos and institution logos?	Use existing four testimonials only if confirmed; hide logo wall otherwise

FINAL GATE

17. Definition of done and launch checklist

#	Check	Pass condition
1	Content approval	Every visible claim is approved:true; no TODO placeholders in production
2	Accessibility	axe zero serious/critical; manual keyboard and screen-reader pass recorded
3	Performance	Lighthouse mobile budgets met on the production URL; JS budget respected
4	Security	Headers verified; npm audit clean at high level; no secrets in repo or bundle
5	Forms	Test lead arrives; spam controls active; consent text approved by legal
6	Legal pages	Privacy and Terms live with approved text; footer links correct
7	SEO	Metadata, OG image, sitemap, robots, JSON-LD validated; previews noindex
8	Analytics	Events firing with no PII; dashboard ready
9	Cross-browser and devices	Chrome, Safari, Firefox, Edge; iOS and low-end Android; 360 to 1440px
10	Operations	Monitoring, rollback, domain and redirects configured; README documents setup, scripts and content editing

Honest-copy rules (apply to every string)

- Say 'helps decide what to practise next', not 'knows exactly what you need'. Say 'personalised', not 'guaranteed results'.
- No ranks, score-improvement figures, pass-rate numbers, 'enterprise-grade', 'certified', 'compliant', or 'bank-level security' unless the team supplies evidence and approves.
- The demo and any dashboard images must carry 'Illustrative'. Testimonials appear exactly as approved, with role and platform.
- No urgency tricks, countdowns, fake scarcity or pre-ticked consent boxes. Calm, clear language suits parents and institutions, especially in a market where trust in edtech has been shaken.

WHERE THIS COMES FROM

18. References and verification notes

- BYOL Academy competitive analysis report (23 October 2026): current-site audit, audience analysis, competitor landscape, wireframe and positioning. Facts about byolacademy.com come from the homepage and About page as read on 23 October 2026.
- Standards to consult while building: WCAG 2.2 (W3C), WAI-ARIA Authoring Practices, Core Web Vitals thresholds (web.dev), Next.js documentation (App Router, Metadata, Route Handlers, Security headers), Tailwind CSS documentation, shadcn/ui documentation, Radix UI, Zod, React Hook Form, Playwright, axe-core, Lighthouse CI, Resend and Upstash docs.
- Bayesian Knowledge Tracing: Corbett and Anderson (1995), the classic model referenced on the About page as a candidate research method. The demo uses it only as an illustration.
- Versions, CLI flags, configuration syntax and plan limits change quickly. Where this guide says 'verify', check current official documentation before relying on it.
- Legal note: this guide is not legal advice. Have qualified counsel review privacy, consent and terms content (including obligations under India's DPDP Act, 2023 and its rules) before launch.